

● MEDIUM

● CRAFT AND STRUCTURE

Craft and Structure

10 questions · ~15 min

04

Text 1

Little is known about how plate tectonics—wherein slabs of Earth’s crust move over, under, away from, and against one another—began. Some researchers contend that tectonic movements began around 3 billion years ago, often noting that computer models of Earth’s mantle temperature at the time indicate that the mantle would have been sufficiently molten to enable the plates to move.

Text 2

Ultimately, any plausible claim about the inception of tectonic movement must rest on empirical evidence from the geological record. Researcher Wriju Chowdhury and his team analyzed the geochemistry of zircon crystals to gain insight into the chemical composition of the magma from which the crystals formed and, based on the data, compellingly argue that plate tectonics may have been occurring as early as 4.2 billion years ago.

Based on the texts, how would the author of Text 2 most likely respond to what “some researchers contend” as described in Text 1?

- A. By suggesting that the temperature of Earth’s mantle 3 billion years ago was likely insufficient to allow for the level of tectonic movement predicted by computer models
- B. By distinguishing between computer models of Earth’s mantle temperature that reliably predict the onset of plate tectonics and those that do not
- C. By indicating that computer models of Earth’s mantle temperature are still being improved such that new models tend to be much more reliable than their predecessors

- D. By asserting that a more definitive form of evidence than the computer models suggests a different timeline for the onset of plate tectonics on Earth

Generally it takes Tule geese about four days to migrate south for the winter. From their summer breeding grounds in Cook Inlet, Alaska, the birds begin by flying over the Gulf of Alaska, keeping about 100 miles from the Canadian shore. They pause to rest on the Pacific Ocean, then fly toward Summer Lake, Oregon, before finally arriving at their winter destination of Sacramento Valley, California. In 2020, however, it took the geese over twice as long to make their way from Cook Inlet to Sacramento Valley. According to researchers, the reason was airborne pollutants.

Which choice best states the function of the underlined portion in the text as a whole?

- A. It illustrates a change in Tule geese's usual flight behavior.
- B. It suggests an explanation for why Tule geese breed in Alaska.
- C. It describes part of the Tule geese's typical winter migration journey.
- D. It compares Tule geese to other birds that migrate south for the winter.

People sometimes dismiss a claim if it comes from a source they regard as self-interested, but from a strictly logical perspective, the source of a claim is _____ blank: it has no direct bearing on whether the claim is true.

Which choice completes the text with the most logical and precise word or phrase?

- A. indistinct
- B. irrelevant
- C. indisputable
- D. implicit

Text 1

The live music festival business is growing in event size and genre variety. With so many consumer options, organizers are finding ways to cement festival attendance as a special experience worth sharing. This phenomenon is linked to the growing “experiential economy,” where many find it gratifying to purchase lived experiences. To ensure a profitable event, venues need to consider the overall consumer experience, not just the band lineup.

Text 2

Music festival appearances are becoming a more important part of musicians’ careers. One factor in this shift is the rising use of streaming services that allow access to huge numbers of songs for a monthly fee, subsequently reducing sales of full-length albums. With this shift in consumer behavior, musicians are increasingly dependent on revenue from live performances.

Based on the texts, both authors would most likely agree with which statement?

- A. Consumers are more interested in paying subscription fees to stream music than in attending music festivals in person.
- B. Consumers’ growing interest in purchasing experiences is mostly confined to the music industry.
- C. Changing consumer behaviors are leading to changes in music-related businesses.
- D. The rising consumer demand for live music festivals also generates higher demand for music streaming platforms.

By combining Indigenous and classical music, Cree composer and cellist Cris Derksen creates works that reflect the diverse cultural landscape of Canada. For her album *Orchestral Powwow*, Derksen composed new songs in the style of traditional powwow music that were accompanied by classical arrangements played by an orchestra. But where an orchestra would normally follow the directions of a conductor, the musicians on *Orchestral Powwow* are led by the beat of a powwow drum.

Which choice best states the main purpose of the text?

- A. To examine how Derksen's musical compositions blend cultures
- B. To argue that Derksen should be recognized for creating a new style of music
- C. To describe the difficulties Derksen encountered when producing her album
- D. To establish a contrast between Derksen's classical training and her Cree heritage

In a 2022 study, Nicolás M. Morato and colleagues attempted to resolve a long-standing paradox: although water is critical to most reactions that enable life, during the earliest stages of Earth's history—when most of the planet's surface was covered by oceans—the presence of water likely _____ blank the formation of proteins and other biomolecules that are found in all living organisms.

Which choice completes the text with the most logical and precise word or phrase?

- A. reflected
- B. hampered
- C. imitated
- D. exploited

Text 1

A tiny, unusual fossil in a piece of 99-million-year-old amber is of the extinct species *Oculudentavis khaungraae*. The *O. khaungraae* fossil consists of a rounded skull with a thin snout and a large eye socket. Because these features look like they are avian, or related to birds, researchers initially thought that the fossil might be the smallest avian dinosaur ever found.

Text 2

Paleontologists were excited to discover a second small fossil that is similar to the strange *O. khaungraae* fossil but has part of the lower body along with a birdlike skull. Detailed studies of both fossils revealed several traits that are found in lizards but not in dinosaurs or birds. Therefore, paleontologists think the two creatures were probably unusual lizards, even though the skulls looked avian at first.

Based on the texts, what would the paleontologists in Text 2 most likely say about the researchers' initial thought in Text 1?

- A. It is understandable because the fossil does look like it could be related to birds, even though *O. khaungraae* is probably a lizard.
- B. It is confusing because it isn't clear what caused the researchers to think that *O. khaungraae* might be related to birds.
- C. It is flawed because the researchers mistakenly assumed that *O. khaungraae* must be a lizard.
- D. It is reasonable because the *O. khaungraae* skull is about the same size as the skull of the second fossil but is shaped differently.

Part of the Atacama Desert in Peru has surprisingly rich plant life despite receiving almost no rainfall. Moisture from winter fog sustains plants once they're growing, but the soil's tough crust makes it hard for seeds to germinate in the first place. Local birds that dig nests in the ground seem to be of help: they churn the soil, exposing buried seeds to moisture and nutrients. Indeed, in 2016 Cristina Rengifo Faiffer found that mounds of soil dug up by birds were far more fertile and supported more seedlings than soil in undisturbed areas.

Which choice best describes the function of the underlined portion in the text as a whole?

- A. It elaborates on the idea that the top layer of Atacama Desert soil forms a tough crust.
- B. It describes the process by which seeds are deposited into Atacama Desert soil.
- C. It identifies the reason particular bird species dig nests in Atacama Desert soil.
- D. It explains how certain birds promote seed germination in Atacama Desert soil.

The Illustrated History of San Mateo County, published in 1878, features several illustrations by Grafton Tyler Brown showing the California county's farms, residences, and businesses. It was _____ blank for a business to be featured, since being included suggested that a business was an important part of the community.

Which choice completes the text with the most logical and precise word or phrase?

- A. risky
- B. misleading
- C. advantageous
- D. unremarkable

Text 1

One challenge faced by researchers studying global urbanization is that countries may define urban settlements differently. Many countries define urban settlements based on the number of people living in them. However, countries do not all use the same number; for example, Iceland uses a minimum population of 200, while the Netherlands uses 20,000. Other countries use a combination of population and other factors. This variation makes it difficult for researchers to compare aspects of urbanization in different countries.

Text 2

Recently, a group of six international organizations developed global definitions of common types of settlements. The group developed a new measure called the “degree of urbanization.” This new measure establishes global criteria used to define three types of settlements (cities, towns, and rural areas) and allows researchers to better understand global urbanization rates.

Based on the texts, how would the author of Text 2 most likely respond to the problem presented in Text 1?

- A. By recommending that a specific institution should further investigate the problem
- B. By suggesting that researchers focus on topics besides urbanization
- C. By supplying additional ways in which urbanization research is difficult
- D. By noting that a possible solution to the problem is available